04

Multi-Agent Orchestration

Five patterns for coordinating multiple agents

5 Orchestration Patterns

Start simple. Add complexity only when needed.

Direct LLM Call → Single Agent → Multi-Agent Orchestration

Sequential
A → B → C
Pipeline

Concurrent
A,B,C parallel
→ Aggregator

Group Chat
Shared thread
Debate & reflect

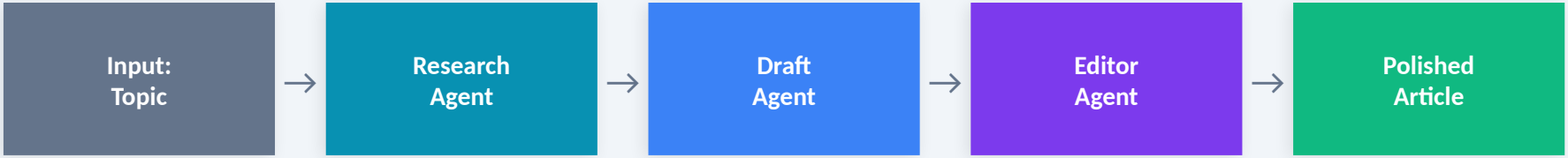
Handoff
Triage → Route
to specialist

Magentic
Manager + Ledger
+ Workers

The biggest practical challenge: how conversation context flows between agents. Each pattern handles this differently — fresh context, shared thread, structured objects, or task-specific context.

Sequential: Content Pipeline

Agents work in a chain — each agent's output becomes the next agent's input.



When to Use

- Clear, ordered stages
- Each stage refines the previous output
- Different persona per stage

Context Strategy: Fresh per Stage

Each agent gets a clean conversation with ONLY the previous agent's output. No 'context pollution' from earlier stages' internal reasoning. Token-efficient since each call is small.

Sequential — Decision Guide

✓ When to Use

Every step depends on the outcome of the previous one

The workflow has a single, linear path

You need tight control and clear ordering

Progressive refinement: draft → review → polish

★ Why Choose It

Predictable, debuggable flow

Great for processes with compliance or risk checks

Ensures no step is skipped

Token-efficient: fresh context per stage

✗ When NOT to Use

Only a few stages that a single agent can accomplish

Agents need to collaborate or see each other's reasoning

Dynamic routing based on intermediate results is needed

Steps can run independently (use Concurrent instead)

➤ Example

Loan Approval Pipeline

Verify applicant info → Evaluate credit → Approve or reject

Exercise: Content Pipeline Code

`exercises/04_sequential/01_content_pipeline.py`

3 Agents, Each with a Focused Prompt

```
research_agent = Agent(
    name="Research Agent",
    system_prompt=RESEARCH_PROMPT, model=model)
draft_agent = Agent(
    name="Draft Writer",
    system_prompt=DRAFT_PROMPT, model=model)
editor_agent = Agent(
```

Fresh Context at Each Stage

```
# Stage 1: Research (own messages)
research_msgs = [{"role": "user",
  "content": f"Research: {TOPIC}"}]
research_out = run(research_agent,
    research_msgs, client)

# Stage 2: Draft (FRESH messages!)
draft_msgs = [{"role": "user",
```

Context Flow: Only Output Passes Forward

Research Agent
Gets: topic string
Produces: research notes

→

Draft Writer
Gets: research notes only
No research agent reasoning

→

Editor Agent
Gets: draft article only
No research or draft reasoning

log_context_pass() makes the handoff visible in console output.

Concurrent — Decision Guide

✓ When to Use

Multiple agents can work on a problem at the same time

Tasks do not depend on one another

You need multiple perspectives (creative, legal, quality)

Speed matters — you want faster throughput

★ Why Choose It

Faster throughput via parallel execution

Each agent provides independent evaluations

Easy to scale horizontally

Ideal for high-volume or time-sensitive workflows

✗ When NOT to Use

Agents need to build on each other's work

Tasks require a specific order or are deterministic

Resource constraints (e.g., model quota limits)

Results require sequential refinement

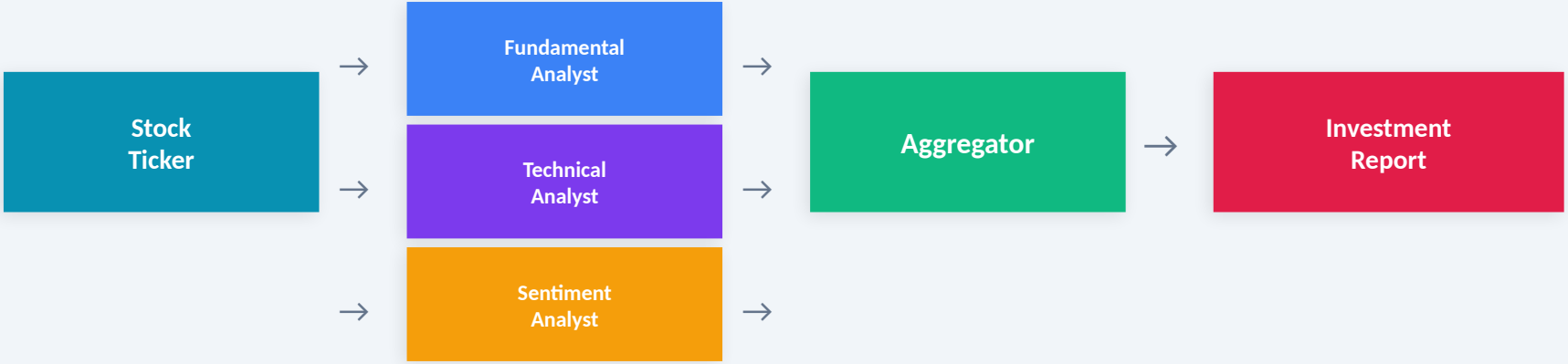
➤ Example

Marketing Campaign Review

Branding + Legal + Compliance + Quality — all review in parallel

Concurrent: Stock Analysis

Multiple agents work on the same input simultaneously, then an aggregator combines results.



When to Use

- Multiple independent analyses of same data
- Each analysis benefits from a different perspective
- Analyses can run without depending on each other

Context Strategy: Isolated per Thread

Each agent gets the SAME initial input independently. No shared state between parallel agents. The aggregator receives ALL outputs combined. Uses ThreadPoolExecutor for real parallelism.

Exercise: Concurrent Stock Analysis Code

exercises/05_concurrent/01_stock_analysis.py

Fan-Out: ThreadPoolExecutor

```
def run_analyst(name, prompt, query, model):
    """Each thread gets its own client!"""
    agent = Agent(name=name,
        system_prompt=prompt, model=model)
    messages = [{"role": "user",
                 "content": query}]
    with get_client() as client:
        output = run(agent, messages, client)
    return name, output
```

Fan-In: Aggregate Results

```
# Combine all outputs into one context
combined = "\n\n".join(
    f"=== {name} ===\n{output}"
    for name, output in results.items()
)

aggregator = Agent(
    name="Aggregator",
    system_prompt=AGGREGATOR_PROMPT,
    model=model)
```

Key Implementation Details

Each thread creates its own get_client() — OpenAI clients are not thread-safe.
as_completed() processes results as they arrive, not in submission order. No shared state between analysts.

Group Chat — Decision Guide

✓ When to Use

Multiple agents need to reason together

Complex problems require back-and-forth refinement

No single agent has all the context or expertise

The final answer emerges from collaboration, not strict order

★ Why Choose It

Higher-quality insights through discussion

Great for ambiguous or exploratory tasks

Agents can challenge and validate each other's ideas

Flexible structure — not locked into sequential or parallel

✗ When NOT to Use

You need strict, step-by-step processing

The problem requires complex dynamic planning

Problem lacks clarity and discussion could go off the rails

Token budget is tight (shared history grows fast)

➤ Example

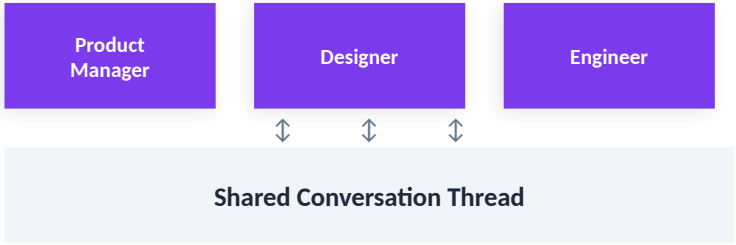
Product Strategy Planning

Engineering + Design + Marketing + Business → Collaborative Alignment

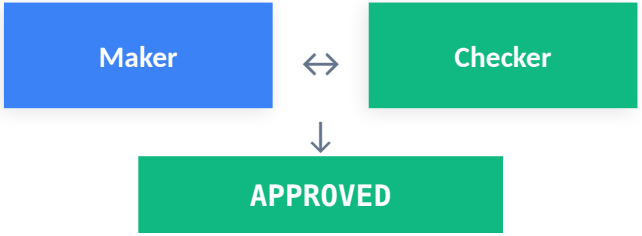
Group Chat: Brainstorm & Maker-Checker

Agents share a conversation and take turns contributing, building on each other's messages.

Brainstorm (Multi-Agent)



Maker-Checker (Two-Agent)



Context Strategy: Shared Messages List

All agents share the SAME conversation thread. Later agents see the full history. Token usage grows with every turn. Maker-checker variant loops until 'APPROVED' keyword appears.

Exercise: Brainstorm — Shared Conversation

exercises/06_group_chat/01_brainstorm.py

3 Agents × 3 Rounds on Shared Thread

```
AGENTS = {
  "Product Manager": "Focus on market fit...",
  "Designer": "Focus on UX, accessibility...",
  "Engineer": "Focus on feasibility...",
}
SPEAKING_ORDER = [
  "Product Manager", "Designer", "Engineer"]
MAX_ROUNDS = 3
```

Round-Robin Speaking Loop

```
for round in range(1, MAX_ROUNDS + 1):
  for agent_name in SPEAKING_ORDER:
    # Inject this agent's persona
    agent_msgs = [
      {"role": "system",
       "content": AGENTS[agent_name]},
      *shared_messages,  # full history
    ]
    response = client.chat.completions
      .create(model=model,
      shared_messages.append(
        {"role": "assistant",
         "content": f"[{agent_name}]: ..."})
```

System prompt is swapped per turn — each agent sees all messages but speaks with its own persona. Name tags [PM]: prefix identify speakers.

Exercise: Maker-Checker — Iterative Refinement

exercises/06_group_chat/02_maker_checker.py

Generate → Review → Revise Loop

```
for iteration in range(1, MAX_ITERATIONS + 1):
    # Review: checker sees shared thread
    review_msgs = [
        {"role": "system", "content": REVIEWER},
        *shared_messages,
        {"role": "user",
         "content": "Review. Reply APPROVED or give feedback."}
    ]
    response = client.chat.completions.create(
        model=model, messages=review_msgs,
        temperature=0.2)  # Low temp for precision
    review = response.choices[0].message.content

    # Check for approval keyword
    if review.strip().upper().startswith("APPROVED"):
        break  # Done!

    # Revise: generator sees feedback
    revise_msgs = [
        {"role": "system", "content": GENERATOR},
        *shared_messages,
        {"role": "user",
         "content": "Address the reviewer's feedback."}
    ]
```

Termination Strategy

- APPROVED keyword in reviewer output
- MAX_ITERATIONS = 4 safety limit
- Low temperature (0.2) for precise review

Reflection Pattern

Also called evaluator-optimizer. The checker provides actionable feedback, the maker addresses it. Both see the full shared_messages history. Quality improves with each iteration.

Chapter 4 Recap: Multi-Agent Orchestration

1. Sequential: fresh context per stage — only output passes forward, no context pollution
2. Concurrent: ThreadPoolExecutor + own client per thread — independent, no shared state
3. Group Chat: shared messages list — all agents see full history (token cost grows)
4. Maker-Checker: APPROVED keyword terminates the loop — quality gate pattern
5. Context strategy is as important as choosing the pattern itself

Next: Chapter 5 — Advanced Patterns → Handoff, Magentic, Production Considerations